

Instructions for Participants: Enterprise Edge Pipeline Activity

Objective: You have to design, check, and integrate your allotted part of an industrial disaster response system. Please assume the role of Lead System Architects for this activity.

I. Material Distribution & Setup

Materials Provided: Each team is given a Sector Task Card, three marker pens (Black, Red, Green), and exactly **three (3) blank A4 sheets**.

How to use the A4 sheets:

- **Paper 1 (Rough Work):** Kindly use this sheet for your initial discussions, rough diagrams, and calculations. This will not be evaluated.
- **Paper 2 (Main Flowchart):** This should be your final, neat architectural diagram. This is your main system logic and will be presented at the end.
- **Paper 3 (Data Transfer Sheet):** Please do not write anything on this until the final stage. This acts as your live "Data Packet". You have to write your final output on this paper and physically hand it over to the next team.

II. Phase 1: Manual System Design (Strictly NO MOBILE PHONES)

In this stage All mobile phones must be kept inside or face-down on the table.

- **Understand your Topic:** Read the Sector Task Card allotted to your team carefully. Understand the specific problem statement and constraints.
- **Draw the Logic (On Paper 2):** Using the **Black Marker**, draw your system's workflow or logic gates and data paths on Paper 2. (Optional) Do not just assume things; you must draw the exact technical steps.
- **Mark Integration Points (On Paper 2):** You have to draw two empty boxes on the edges of Paper 2:
 1. An [INPUT BOX] on the left side (where you will keep the data sheet received from the previous team).
 2. An [OUTPUT BOX] on the right side (where you will write the output for the next team).

Time Allotted: You have exactly **20 minutes** to complete this final diagram.

III. Phase 2: AI Verification (MOBILES / AI TOOLS ALLOWED)

You can use your mobile phones now. You will use AI tools to find out mistakes or vulnerabilities in the logic you just designed.

- **Open AI Tools:** Kindly open ChatGPT, Gemini, or Claude on your phones.
- **Give the Prompt:** Find the specific *AI Verification Prompt* printed at the bottom of your Sector Task Card.
- **Enter your details:** Type that exact prompt into the AI, and then type a brief explanation of the flowchart you drew on Paper 2.
- **Apply the Fix:** Read the AI's feedback. Find out what vulnerability it pointed out. Take your **Red Marker** and draw a "Security Patch" (like a new rule or filter) directly onto Paper 2 to fix that issue. You have **10 minutes** for this.

IV. Phase 3: Live Integration Activity

Now we will test the full system. We will pass one single data packet from the first team to the last team using Paper 3.

- **Initial Step:** Team 1 (Sector 1) will write their initial data payload in big letters using the **Black Marker** on their **Paper 3**.
- **Passing the Data:** One member from Team 1 will walk over and place their Paper 3 on Team 2's table, exactly over their [INPUT BOX].
- **Live Processing:** Team 2 will look at the data. You have 60 seconds to explain how that data flows through your Paper 2 diagram, announce the result loudly to the room, and write your new output onto your own team's **Paper 3** using the **Green Marker**.
- **Forwarding:** Team 2 will then walk and hand over their Paper 3 to Team 3. This cycle continues until the data reaches the final team (Sector 4).
- **Cross-Questioning:** Please be ready to answer practical scenario questions from the facilitator while you are processing this data.

V. Phase 4: Final Presentation & Review

Connecting the entire system.

- **Connecting the Flow:** One representative from each team will bring their **Paper 2** (Main Flowchart) to the front of the room.
- **Combine the Sheets:** The four A4 sheets will be kept side-by-side, slightly overlapping, to show the complete data pipeline from the first sensor up to the cloud.
- **Overall Discussion:** Everyone will participate in a final discussion to understand where the system integration can fail, based on how smoothly the Paper 3 sheets moved from one team to the next.

Referance : Ordered sequentially to demonstrate how data moves through the system.

